

TIP4PATLIBS

Get on the platform. Then ask better questions.

Six modules on the EPO Technology Intelligence Platform. You do not have to learn SQL first — you have to get onto the platform, and then decide what to ask.

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github.com/mtcberlin/epo-tip4patlibs

THE PLATFORM EVERYTHING HERE RUNS ON

What TIP is — and what your PATLIB gets out of it

The EPO Technology Intelligence Platform

tip.epo.org

- A JupyterLab server adapted to and hosted by the EPO
- PATSTAT, the EPO Register and the EP-A/B full-text databases, queryable as SQL
- Free access for the general public
- Extensive training material on database structure, Python and SQL from the EPO Academy

What this opens up for a PATLIB

the reason we spent the year on it

- Patent analytics and reports built on current PATSTAT — not on an export from last year
- A place to get familiar with patent statistics and data-science methods, at your own pace
- No licence to negotiate, no server to run — and what you build stays yours

The entry barrier is an account, not a budget. Everything shown over these two days runs inside TIP.

HOW THIS CAME ABOUT

EPO handed us python and sql courses. Then we walked into TIP together.

Several walking-group sessions with colleagues from the PATLIB community, working out access to the Technology Intelligence Platform side by side — not a product demo, but people trying to get a first query to return rows.

The same questions, every time

What survives a restart. How to reach PATSTAT. Where a query even starts. How to show a result to a colleague who does not code.

Written down as it worked

Each answer became a notebook that runs top to bottom inside TIP — no installation, and a sensible default so it works before you change anything.

Five modules, one repository

In the order we found them. Everything shown over these two days is in there, public, yours to run and to change.

Public repository available to you: <https://github.com/mtcberlin/epo-tip4patlibs>



BRING IT INTO YOUR OWN TIP

One command. Then you have everything.

Open <https://tip.epo.org>, log in with your EPO account, start a terminal (File → New → Terminal) and run this one line:

TIP TERMINAL

```
curl -fsSL https://tip.depa.tech/install | bash
```

The five modules

Cloned into ~/epo-tip4patlibs, executed, with the code folded away. Open a folder and read.

An assistant on standby

Installed and waiting. Sign in only when you want to change a notebook or build your own analysis.

It survives the rebuild

TIP rebuilds your machine without warning. Run the same line again: it repairs the setup and updates the material.

ONE WORD YOU WILL HEAR ALL DAY

PATSTAT — and who writes the queries

1 PATSTAT

the EPO's raw patent database

Not a search interface. Around forty linked tables holding every application, applicant, classification, family link and citation the offices have published.

There is no search box and no company record. You ask it questions in SQL — which is exactly why it stayed a specialist tool for twenty years.

2 Who writes the SQL

three routes, and you will see all three

The EPO Academy material on TIP, which teaches the schema and the language properly — the long road, and the one worth walking.

A query library that already runs, so your first result arrives before your first lesson does. Or an assistant in your session that drafts it for you.

Whichever route: the query stays on the screen, and you are the one who reads it.

Nothing in these two days asks you to write SQL from scratch. Everything in them asks you to read a result critically.

The barrier moved.

For twenty years the hard part of patent analytics was getting to the data at all: the licence, the schema, the SQL. On TIP, none of those is where your time goes any more. What is left is the part that was always the professional work.

What no longer costs you weeks

- Getting access to current PATSTAT at all
- Knowing which of the forty tables to join
- Writing and debugging the SQL by hand
- The distance between a question and a first table

What stays yours

- Deciding what is actually being asked
- Defining the corpus — and its exclusions
- Judging whether the answer holds
- The caveat you put under the table

Six modules follow. Every one of them is a live notebook in TIP — the slides only set it up.

THE RUNNING ORDER

One claim, three examples, **two use cases.**

Rising ambition from left to right. Each module is set up on two slides, then shown live in TIP.

THE CLAIM

1

Start with TIP
install an assistant that
survives the restart

THREE EXAMPLES

2

Query Library
a question you are actually asked

3

PATSTAT Explorer
the same query, as an app

4

Regional Analysis
your region, your client list

TWO USE CASES

5

Antibiotics Landscape
a multi-step analysis, end to end

6

IPScore with TIP
what is this patent worth?

Modules 1–4 Arne Krüger · mtc.berlin

Modules 5–6 Riccardo Priore · Centro PATLIB, AREA Science Park

0

A setup that survives the restart

`install.sh`

“You have a login to a machine you do not own, which is rebuilt without warning, and which holds the largest patent database in the world. How do you get a working setup onto it that is still there next session?”

WHY THE OBVIOUS ANSWER FAILS

“I will just install what I need.” The usual install and configuration is gone by Monday — nothing was deleted, the machine was simply rebuilt.

“Then I will work through the Academy material first.” Do — it is good. But you can have a result on the screen long before you are finished.

“I would need a developer for this.” Not for this. You need a setup that persists, and the habit of reading the query before you trust the number.

LIVE IN TIP NEXT

The setup surviving a restart, then a first PATSTAT query — company and institution search — read line by line before we trust the number.

0

Setup — what you have at the end

`install.sh`

WHAT NOW EXISTS

- A TIP working environment that is still there after the machine restarts
- One sample query you can read, run and change — before you could have written it
- A setup you can carry into your own PATLIB's TIP account
- AI Coding Assistant can be installed optionally to support you in your learning and implementing your ideas.

THE ONE SENTENCE TO REMEMBER

The barrier was never the database. It was the SQL and the PATSTAT knowledge — and that is the part that has stopped standing in your way.

Full 45-minute version of this block: [9_misc/handouts/](#)

1 The Query Library

1_querylib/

“A client walks in and asks: who in Europe is working on solid-state batteries?”

WHY THE OBVIOUS ANSWER FAILS

The first answer was always a keyword search. It is not wrong — it is simply the weakest instrument you have.

Keywords miss everything phrased differently, and catch everything that merely mentions the term in passing.

Neither problem is visible in the result. You get a list, and no way to show that it is complete.

LIVE IN TIP NEXT

One query from the library run against PATSTAT Global, then the same query with a single parameter changed.

1

Module 1 — what you have at the end

1_querylib/

WHAT NOW EXISTS

- One query adapted to a question you are actually asked, with its result table
- A first feel for what a PATSTAT query costs in time
- The vocabulary to explain why keywords were combined with classifications

THE ONE SENTENCE TO REMEMBER

An analysis you cannot understand is
one you cannot defend.

Full 45-minute version of this block: [9_misc/handouts/](#)

2

PATSTAT Explorer

2_patstat_explorer/

“A client asks how big Siemens Healthineers' patent portfolio is. You run a name search. You get 200 rows. Which one is the answer?”

WHY THE OBVIOUS ANSWER FAILS

None of them. PATSTAT does not know companies — it knows names, exactly as typed on each application.

One entry per spelling, per country, per subsidiary, per typo. There is no company record to look up.

Take the largest row and you have undercounted the group — as a clean, confident-looking number.

LIVE IN TIP NEXT

A name search returning 200 rows, the consolidation decision behind the number, and the same search running as an app inside TIP.

2

Module 2 — what you have at the end

2_patstat_explorer/

WHAT NOW EXISTS

- One consolidated applicant group — a named list of spellings you decided belong together
- The group's family count, filing offices and technology profile
- A written note of what your consolidation missed

THE ONE SENTENCE TO REMEMBER

Consolidation is a judgement, not a lookup. What makes the number defensible is that you wrote the decision down.

Full 45-minute version of this block: [9_misc/handouts/](#)

3 Regional Lead Generation

3_lead_generation/

“Your PATLIB is asked to justify its outreach budget. Which companies in your region should you actually be talking to — and why those?”

WHY THE OBVIOUS ANSWER FAILS

Every PATLIB has a contact list. Almost none of them can say how it was built.

Who walked in, who came to the last event, who somebody happened to remember.

The question a ranked list has to survive is not “who is on it” but “who is missing”.

LIVE IN TIP NEXT

The regional notebook on Alsace (FR42), then the same run for a German region — and the tier grid it produces.

3

Module 3 — what you have at the end

3_lead_generation/

WHAT NOW EXISTS

- A named shortlist for your region — family count, depth tier, reach tier, zones
- The grid showing how your region distributes across the tiers
- One defensible sentence about what the list does not contain

THE ONE SENTENCE TO REMEMBER

This is the EP/PCT-active subset of your region. Roughly 70% of national families never take that route — say so under the table.

Full 45-minute version of this block: [9_misc/handouts/](#)

4

Patent landscape reports

4_patentreports/

Landscape analyses after Riccardo Priore, Centro PATLIB, AREA Science Park

“A regional cluster manager asks for a picture of what is happening in antibiotic resistance. Where do you start?”

WHY THE OBVIOUS ANSWER FAILS

“Drug resistance” is dominated by cancer — an oncology landscape wearing the wrong label.

Bare acronyms — MRSA, VRE, ESBL — are noisy, and turn up where the subject is not.

Neither is visible in the result: a large corpus, looking perfectly healthy, about something else.

LIVE IN TIP NEXT

The assembled report with its 13 charts — then back to the search strategy that defined the corpus.

4

Module 4 — what you have at the end

4_patentreports/

WHAT NOW EXISTS

- One self-contained HTML report, 13 charts, no internet connection needed
- One workbook carrying the data behind every chart
- A written search strategy — including its exclusions

THE ONE SENTENCE TO REMEMBER

The analyses are generic. The corpus definition is the part that is yours, and the part you have to defend.

Full 45-minute version of this block: [9_misc/handouts/](#)

5

What is this patent worth?

5_ipscore/

IPScore is an EPO tool; this is the course's own implementation of the 3.01 model

“A client puts a granted European patent on the table and asks what it is worth.”

WHY THE OBVIOUS ANSWER FAILS

Renewal costs, comparable licence rates, “it depends” — or the answer that the question cannot be answered.

IPScore does answer it: 40 questions, seven figures from the accounts, a ten-year discounted cash flow.

And then: all 40 answers come from a person. The engine is exact; its inputs are opinions.

LIVE IN TIP NEXT

The questionnaire and the NPV for one granted European patent, and which of the 40 answers the data could check.

5

Module 5 — what you have at the end

5_ipscore/

WHAT NOW EXISTS

- A verified implementation of the EPO IPScore model, seen from the inside
- One completed valuation of a real patent — NPV and profile
- A statement of how much of that number is actually evidence

THE ONE SENTENCE TO REMEMBER

None of the 11 questions PATSTAT can check is one of the 8 that carry money. What a database can check and what a valuation depends on barely overlap.

Full 45-minute version of this block: [9_misc/handouts/](#)

The demo is the repository.

1

A working environment on TIP that survives the restart

free, hosted, with current PATSTAT behind it

2

Three working examples, in rising order of ambition

a ready question · the same query as an app · a named shortlist for your own region

3

Two use cases built the same way

a landscape report, and a patent valuation

4

And in every one of them, the part no tool does for you

deciding what to ask, and judging whether the answer holds



Arne Krüger on LinkedIn

Seven handouts — the full 45-minute version of every block you just saw — in `9_misc/handouts/`
`github.com/mtcberlin/epo-tip4patlibs`